

The Stability of Stochastic Coupled Systems With Time-Varying Coupling and General Topology Structure

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Abstract—We introduce a class of novel stochastic coupled systems in which the coupling structure is time-varying and the topology structure is not strongly connected, and first establish the system on a digraph with a time-varying weight matrix. Motivated by Du and Li (2014), we give a hierarchical method to deal with digraphs without strong connectivity and establish the corresponding hierarchical algorithm to realize this approach. Also, an example is given to illustrate our hierarchical algorithm and its feasibility. In the sequel, based on the theory of asymptotically autonomous systems, Kirchhoff's matrix tree theorem, and Lyapunov method, several moment exponential stability criteria are presented, including a Lyapunov-type theorem and a coefficient-type criterion. Furthermore, theoretical results are applied to stochastic coupled oscillators with time-varying coupling structure (SCTCS), and the stability criterion of SCTCS is obtained. Finally, the effectiveness of theoretical results is illustrated by two numerical examples.

Index Terms—Connectivity, exponential stability, stochastic coupled oscillators, time-varying coupling structure.

I. INTRODUCTION

DURING the past few decades, studies about coupled systems are crucial, motivated by diverse applications in mechanics, mathematical epidemics, electrical engineering, and biological systems [1]–[7]. In the real world, random disturbance and time delays are inevitable. Therefore, stochastic coupled systems [8]–[10] and some kinds of delay systems [11]–[15] are investigated extensively. In this paper, we consider stochastic coupled systems with time-varying coupling and general topology structure (SCTGS)

as follows:

$$\begin{aligned} dx_k(t) = & f_k(x_k(t), x_k(t - \tau), t)dt \\ & + \sum_{h=1}^l a_{kh}(t) H_{kh}(x_k(t), x_h(t))dt \\ & + g_k(x_k(t), x_k(t - \tau), t)dB(t), \quad k = 1, 2, \dots, l \end{aligned} \quad (1)$$

where $a_{kh}(t)$ represents the time-varying coupling strength and H_{kh} stands for the coupling function. Here, we emphasize that SCTGS are a class of novel stochastic coupled systems. The majority of literatures assume that the coupling structure of coupled systems is constant, and that some studies about coupled systems with a constant coupling structure (i.e., $a_{kh}(t) = a_{kh}$) can be found in [16]–[19]. Li *et al.* [20] investigated the global exponential stability of stochastic coupled oscillators. Toshikazu and Yoshiaki [21] studied the global stability of a multigroup SIS epidemic model. However, in mechanical systems and epidemic models, the weight of state transfer among coupled oscillators and the diffusion rate of epidemic always change over time. Therefore, to meet practical applications, it is of great significance to study coupled systems with a time-varying coupling structure.

The stability of coupled systems is an active and rich research area, with applications to ecosystems, epidemic models, and physics. Therefore, many researchers have investigated the stability of coupled systems by different approaches, such as the Lyapunov method [12], the linear matrix inequalities approach [22], [23], [25], [39], and various kinds of control methods like adaptive control [26]–[28] and impulsive control [29]. Guo *et al.* [18] and Li and Shuai [19] investigated the stability of coupled systems by establishing a system on a digraph and presented an effective method that is combining the graph theory with the Lyapunov method. Compared with methods provided in pervious literatures, the method overcomes the difficulty of straightly constructing an appropriate Lyapunov function. Motivated by Zhang *et al.* [30], [32] and Guo *et al.* [31], some researchers have used the method to study the dynamics of stochastic coupled systems, multigroup models, and coupled reaction diffusion systems, and many related results have been reported.

We find that the assumption of results in literatures mentioned above requires digraphs to be strongly connected.

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However, digraphs with strong connectivity are a special case. In the real world, many systems are established on digraphs without strong connectivity (DWSCs), and DWSCs exist in different fields. For example, in social sciences, the crowd through social interaction makes up social graphs. In social graphs, one person represents a vertex, and degrees of acquaintance among each other stand for weights of social graphs. In ecological systems, epidemic models can be established on DWSCs. Therefore, DWSCs exist in life everywhere and it is of great significance to investigate them. As far as we know, there are a few researchers considering the case of DWSCs.

On DWSCs, the connection among nodes is irreversible, which makes the function and position of different nodes different. Therefore, it is important to classify the nodes of DWSCs. For example, family trees can be seen as DWSCs, and every node represents a family member. The nodes of family trees can be classified on the basis of nodes belonging to different generations. This implies that the classification for nodes is an efficient method to deal with DWSCs. Du and Li [33] investigated the impact of the network connectivity on the synchronization and global dynamics of coupled systems of differential equations. They pointed out that a condensed digraph can be obtained by compressing each strongly connected component into a vertex. It is more interesting that a condensed digraph can be layered by a hierarchical method, so we give a theoretical framework to state the method. Furthermore, a hierarchical algorithm can be established to achieve the hierarchical method.

In this paper, a class of novel stochastic coupled systems SCTGS are presented and the stability of SCTGS on a DWSC is investigated. Next, a hierarchical method for a condensed digraph is given and a hierarchical algorithm is established accordingly. Following the hierarchical algorithm, we can layer a condensed digraph into different layers according to its topology structure, of which every layer consists of strongly connected components. By the theory of asymptotically autonomous systems [36], [37], Kirchhoff's matrix tree theorem, and Lyapunov method on every strongly connected component, several moment exponential stability criteria are presented, including a Lyapunov-type theorem and a sufficient criterion in the form of coefficients. As an application, the theoretical results are applied to stochastic coupled oscillators with a time-varying coupling structure (SCTCS) and two numerical examples are presented to illustrate the effectiveness of the proposed criteria.

The chief contributions are threefold as follows.

- 1) We present a class of novel stochastic coupled systems, namely SCTGS. The coupling structure is time-varying and the topology structure is not strongly connected.
- 2) We give a hierarchical method to deal with DWSCs and a hierarchical algorithm to realize this approach.
- 3) We combine the theory of asymptotically autonomous systems, Kirchhoff's matrix tree theorem, and Lyapunov method to investigate the stability of SCTGS.

The organization of this paper is as follows. In Section II, a hierarchical method and the corresponding hierarchical algorithm are introduced. Then, Section III is devoted to the stability for SCTGS and sufficient criteria of moment

exponential stability. In Section IV, the stability criterion for SCTCS is provided. Numerical examples about SCTCS and stochastic recurrent neural networks are given in Section V. Finally, an example is provided to display the feasibility of the hierarchical algorithm in the Appendix.

Notations: Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a complete probability space with a filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions, and $B(t)$ be a 1-D Brownian motion defined on the space. The mathematical expectation with respect to the given probability measure $\mathbb{P}$ is denoted by $\mathbb{E}(\cdot)$. Write $|\cdot|$ for the Euclidean norm for vectors or the trace norm of matrices. Let $\mathbb{R}^n$ stand for the n -dimensional Euclidean space. The superscript "T" stands for the transpose of a vector or a matrix. Let $C([-\tau, 0]; \mathbb{R}^n)$ be the space of continuous functions $x : [-\tau, 0] \rightarrow \mathbb{R}^n$ with norm $\|x\| = \sup_{-\tau \leq u \leq 0} |x(u)|$. Let $L^2_{\mathcal{F}_0}([-\tau, 0]; \mathbb{R}^n)$ stand for the family of $\mathcal{F}_0$ -measurable $C([-\tau, 0]; \mathbb{R}^n)$ -value random variables y such that $\mathbb{E} \|y\|^2 < \infty$. The notations $\mathbb{L} = \{1, 2, \dots, l\}$, $\mathbb{Z}^+ = \{1, 2, \dots\}$, $\mathbb{R}^+_+ = [0, +\infty)$ and $m = \sum_{i=1}^l m_i$ for $m_i \in \mathbb{Z}^+$ are used. And $C^{2,1}(\mathbb{R}^n \times \mathbb{R}^+_+; \mathbb{R}^+_+)$ represents for the family of all nonnegative functions $V(x, t)$ on $\mathbb{R}^n \times \mathbb{R}^+_+$, which are continuously twice differentiable in x and once in t .

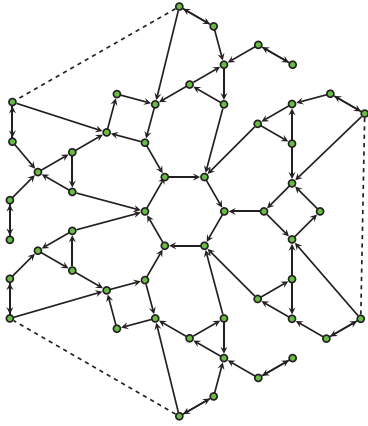
II. HIERARCHICAL METHOD FOR DWSCS

To cope with DWSCs, in this section, a hierarchical method for a condensed digraph is given and a hierarchical algorithm is established accordingly. First, we introduce some concepts of graph theory. Let $(\mathcal{G}, A)$ be a weighted digraph with l ($l \geq 2$) vertices, where $A = (a_{ij})_{l \times l}$ is the weight matrix. Set $V(\mathcal{G})$ as the set of vertex in $\mathcal{G}$. In this paper, $V(\mathcal{G}) = \mathbb{L}$. The Laplacian matrix of $(\mathcal{G}, A)$ is defined as $L = (p_{kh})_{l \times l}$, where $p_{kh} = -a_{kh}$ for $k \neq h$ and $p_{kh} = \sum_{j \neq k} a_{kj}$ for $k = h$. More details can be found in [19]. Similarly, we can define $(\mathcal{G}, A(t))$ as a digraph with time-varying weight matrix $A(t)$ and other notations are defined like above notations accordingly.

A. Hierarchical Method

First, we introduce some concepts about the partial order in [33]. The relation among vertices of a digraph $\mathcal{G}$ as a partial order $\leq$ is defined. For vertices $k, h \in V(\mathcal{G})$, $k \leq h$ if there is a directed path from vertex k to vertex h , where $V(\mathcal{G})$ represents the vertex set of a digraph $\mathcal{G}$.

Second, we define a strongly connected component R of a digraph $\mathcal{G}$ as follows: if a subgraph R is strongly connected, and for any vertex k , a subgraph consisting of the vertex set $V(R) \cup \{k\}$ is not strongly connected, and then R is a strongly connected component. We compress each strongly connected component of the digraph $\mathcal{G}$ into a single vertex, all of which constitute a new digraph $\mathcal{R}$ treated as a condensed graph of $\mathcal{G}$. For a condensed digraph $\mathcal{R}$, a strict partial order $<$ is also defined. Concretely, for $R, \bar{R} \in V(\mathcal{R})$, $R < \bar{R}$ if there are vertex $k \in R$ and vertex $h \in \bar{R}$ such that $k \leq h$ in the digraph $\mathcal{G}$. By means of the definition of a strongly connected component, we can obtain that if $R, \bar{R} \in V(\mathcal{R})$ satisfy $R < \bar{R}$ and $\bar{R} < R$, then R and $\bar{R}$ are the identical strongly connected component. Therefore, relation $<$ is a strict partial order. Furthermore, for a condensed digraph $\mathcal{R}$ with

Fig. 1. Digraph $(\mathcal{G}, A(t))$ with 54 vertices.

a finite number of vertices, minimal and maximal elements exist in $V(\mathcal{R})$ with respect to the strict partial order $\prec$.

Motivated by [33], a hierarchical method is given to layer a condensed digraph. Before that, a hypothesis is imposed first, which ensures that the connectivity of digraphs is constant along with the change of time. Weights in digraphs with a time-varying coupling structure are assumed to satisfy the following assumption in this paper.

Assumption 1: For a given digraph $(\mathcal{G}, A(t))$, assume that $a_{kh}(t)$ satisfies the following conditions: if there exists a $t_0 \geq 0$ such that $a_{kh}(t_0) = 0$, then $a_{kh}(t) \equiv 0$ for any $t \geq 0$. Otherwise, $a_{kh}(t) > 0$ for any $t \geq 0$.

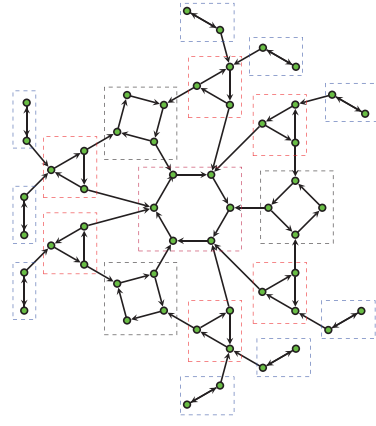
As for a condensed digraph $\mathcal{R}$, we propose a hierarchical concept and each vertex will belong to a certain layer. It should be pointed out that each vertex of a condensed digraph $\mathcal{R}$ is a strongly connected component of a digraph $(\mathcal{G}, \tilde{A}(t))$. In the sequel, a hierarchical method is given in detail.

In a condensed digraph $\mathcal{R}$, all vertices with zero in-degree belong to the first layer. Vertex U is called the vertex in k th ($k \geq 2$) layer, if and only if the following conditions hold.

- 1) There exists a vertex V belonging to the $(k-1)$ th layer at least satisfying that $V \prec U$.
- 2) If there exists a vertex W such that $W \prec U$, then the sequence number of the layer that vertex W belongs to will not exceed $k-1$.

For the purpose of showing the hierarchical method given in this paper clearly and how to layer a DWSC in detail, an example is shown as follows.

Example 1: Given a digraph $(\mathcal{G}, A(t))$ with 54 vertices, which is not strongly connected as shown in Fig. 1 and $(\mathcal{G}, A(t))$ satisfies Assumption 1. Suppose that dotted lines in Fig. 1 satisfies $\lim_{t \rightarrow \infty} a_{kh}(t) = 0$, then the arcs are removed. The corresponding condensed digraph $\mathcal{R}$ is obtained as shown in Fig. 2. We layer the condensed digraph $\mathcal{R}$ into four layers and each rectangle represents a strongly connected component. Blue dashed rectangles belong to the first layer owing to their zero in-degree. Red dashed rectangles are part of the second layer on account of that they are only influenced by the first layer. The third layer influenced by the second layer consists of black dashed rectangles. The purple dashed

Fig. 2. Digraph $(\mathcal{G}, \tilde{A}(t))$ and the corresponding condensed digraph $\mathcal{R}$.

rectangle is in the fourth layer, because it is affected by the third layer.

Remark 1: If a digraph $(\mathcal{G}, \tilde{A}(t))$ is strongly connected, then a digraph $(\mathcal{G}, A(t))$ only has one strongly connected component, which implies that a condensed digraph $\mathcal{R}$ has only a vertex. When $(\mathcal{G}, \tilde{A}(t))$ is not strongly connected, a condensed digraph $\mathcal{R}$ can be layered (see Example 1).

B. Hierarchical Algorithm

It is difficult to layer a digraph with a large number of vertices artificially; however, to the best of our knowledge, a few researchers can give a systematic method to deal with this kind of digraphs. Therefore, we propose a hierarchical algorithm by employing Tarjan's strongly connected component algorithm [34]. In detail, the hierarchical algorithm is presented as follows.

- 1) Give the adjacency matrix J of digraph $(\mathcal{G}, \tilde{A}(t))$. By Assumption 1, the connectivity of $(\mathcal{G}, \tilde{A}(t))$ is constant. Therefore, the adjacency matrix J is constant.
- 2) Employ Tarjan's strongly connected component algorithm to obtain all strongly connected components of $(\mathcal{G}, \tilde{A}(t))$ with order. Thus, we can get an array of sequential vertices number defined as $Array1[]$. Also the number of vertices in each strongly connected component can form an array with the corresponding order defined as $Array2[]$.
- 3) Adjust the matrix J in the order of vertices according to $Array1[]$ and the new matrix is defined as J' .
- 4) Divide J' by employing the function of $mat2cell()$ [$mat2cell()$ in MATLAB is a block function] and the corresponding parameters are $(J', Array2[], Array2[])$.
- 5) Dispose each block of the adjacency matrix J' in the following way.
 - a) If all the elements of a block are 0, then replace this block with 0.
 - b) If not all the elements of a block are 0, then replace this block with 1.
 - c) Set all diagonal elements as 0.

Afterward, a new matrix is obtained defined as J'' and this is the adjacency matrix of the condensed digraph $\mathcal{R}$.

- 6) Layer the condensed digraph as follows: search matrix J'' to seek out those vertices whose corresponding rows in matrix J'' are all zero. Undoubtedly, the in-degree of these vertices is zero and these vertices belong to the first layer. Then drop these vertices and we get a new matrix. Repeat the procedure mentioned above to obtain vertices that belong to the second layer, third layer, and all the layers.

In order to display this hierarchical algorithm preferably, take a DWSC with six vertices for example, which is shown in the Appendix.

In the above-mentioned descriptions, a question arises naturally whether the matrix obtained in step 4) is the adjacency matrix of the condensed digraph $\mathcal{R}$. In order to answer this question, Theorem 1 is given.

Theorem 1: After that, we divide the matrix J into $J' = (J_{ij})_{m \times m}$ by step 4), where m is the number of the strongly connected component R_i . For $i = j$, we define $e_{ii} = 0$, and otherwise

$$e_{ij} = \begin{cases} 0, & \text{if all the elements of } J_{ij} \text{ are 0} \\ 1, & \text{otherwise} \end{cases}$$

then $J'' = (e_{ij})_{m \times m}$ is the adjacency matrix of the corresponding condensed digraph $\mathcal{R}$.

Proof: On the one hand, if not all elements of J_{ij} are 0, since $i \neq j$, we can conclude that for some vertex $k \in V(R_j)$, there exists vertex $h \in V(R_i)$, such that $h \leq k$, which implies $R_j < R_i$. Hence, $e_{ij} = 1$ in J'' . On the other hand, for the case that all the elements of J_{ij} are 0, it is required to prove $e_{ij} = 0$ in J'' . Suppose that $e_{ij} = 1$, then we can show $R_j < R_i$. It means that there are vertex $k \in V(R_j)$ and vertex $h \in V(R_i)$ satisfying $k \leq h$, which is in contradiction with the fact that all the elements of J_{ij} are 0. Thus, $e_{ij} = 0$ in this case. This completes the proof. ■

Remark 2: Recently, the dynamics of systems on a strongly connected digraph has been investigated in [18]–[20]. As far as we know, a few researchers devote themselves to the research of systems on a DWSC, which is a complex and formidable task. Our hierarchical method takes full advantages of the topology relationship among strongly connected components. Thus, our hierarchical method is a powerful tool to handle this task, and the hierarchical algorithm is easy to be implemented by the mathematical software.

III. MOMENT EXPONENTIAL STABILITY FOR SCTGS

At the beginning of this section, we assume that functions f_k , g_k , and H_{kh} for $k, h \in \mathbb{L}$ are continuous and satisfy the local Lipschitz condition and the linear growth condition. In view of [35, Ch. 5, Th. 2.5], it is easy to see that system (1) has a unique solution $x(t) = (x_1^T(t), x_2^T(t), \dots, x_l^T(t))^T$ with initial condition $x(t) = \phi(t)$, $t \in [-\tau, 0]$. We further suppose that $f_k(0, 0, t) = 0$, $g_k(0, 0, t) = 0$ and $H_{kh}(0, 0) = 0$, and then system (1) has a trivial solution $x(t) \equiv 0$. Moreover, suppose that $\lim_{t \rightarrow \infty} f_k(x_k, \tilde{x}_k, t) = \tilde{f}_k(x_k, \tilde{x}_k)$, $\lim_{t \rightarrow \infty} g_k(x_k, \tilde{x}_k, t) = \tilde{g}_k(x_k, \tilde{x}_k)$, and $\tilde{f}_k$ and $\tilde{g}_k$ satisfy the local Lipschitz condition.

Then, two useful definitions about the p th moment exponential stability and the vertex Lyapunov function are recalled and an assumption is given. Define the i th strongly connected component belonging to the j th layer as R_{ji} .

Definition 1: The trivial solution of system (1) is said to be the p th moment exponentially stable (ME-stable) if the p th moment Lyapunov exponent

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \ln(\mathbb{E}|x(t, \phi)|^p) < 0$$

for some $p > 0$ and all $\phi \in L^2_{\mathcal{F}_0}([-\tau, 0]; \mathbb{R}^m)$. When $p = 2$, it is said to be exponentially stable in mean square.

Definition 2: For any $k \in V(R_{ji})$, function $V_k(x_k, t) \in C^{2,1}(\mathbb{R}^{m_k} \times \mathbb{R}_+^1; \mathbb{R}_+^1)$ is called the k th vertex Lyapunov function if the following conditions are satisfied.

V1: There exist positive constants $p \geq 2$, α_k , and β_k , such that

$$\alpha_k |x_k|^p \leq V_k(x_k, t) \leq \beta_k |x_k|^p. \quad (2)$$

V2: There exist positive constants σ_k and η_k , a function $P_k(x_k)$, and a matrix $D = (d_{kh})_{l \times l}$, $d_{kh} \geq 0$, such that

$$\begin{aligned} \mathcal{L}V_k(x_k, t) \leq & \sum_{h \in V(R_{ji})} d_{kh} [P_h(x_h) - P_k(x_k)] \\ & - \sigma_k |x_k|^p + \eta_k |x_k(t - \tau)|^p \end{aligned} \quad (3)$$

for all $x_k \in \mathbb{R}^{m_k}$, $x_h \in \mathbb{R}^{m_h}$.

Assumption 2: Functions $H_{kh}(x_k, 0) = 0$, for all $x_k \in \mathbb{R}^{m_k}$ and $k \in \mathbb{L}$.

After that, we establish system (1) on a DWSC. Then, system (1) restricted on R_{ji} is established as follows:

$$\begin{aligned} dx_k(t) = & f_k(x_k(t), x_k(t - \tau), t)dt \\ & + \sum_{h \in V(R_{ji})} b_{kh}(t) H_{kh}(x_k(t), x_h(t))dt \\ & + \sum_{h \in \bigcup_{j_1 < j} \bigcup_{i_1} V(R_{j_1 i_1})} c_{kh}(t) H_{kh}(x_k(t), x_h(t))dt \\ & + g_k(x_k(t), x_k(t - \tau), t)dB(t), k \in V(R_{ji}) \end{aligned} \quad (4)$$

where $f_k, g_k : \mathbb{R}^{m_k} \times \mathbb{R}^{m_k} \times \mathbb{R}_+^1 \rightarrow \mathbb{R}^{m_k}$. Then, an assumption is needed, which can show a close relationship between the topology structure and the dynamics.

Assumption 3: Functions $c_{kh}(t)$ are bounded, where $c_{kh}(t)$ represent weights among layers. Functions $b_{kh}(t)$ satisfy $b_{kh}(t) \leq 0$ and stand for weights in a same strongly connected component.

Next, a Lyapunov-type theorem is given to investigate the stability of the trivial solution of system (1).

A. Lyapunov-Type Theorem

Theorem 2: Assume that Assumptions 1–3 are satisfied, and system (4) admits vertex Lyapunov functions $V_k(x_k, t)$ for any $k \in \mathbb{L}$. Then the trivial solution of system (1) is ME-stable, if

$$-\sigma_k + \eta_k < 0, \quad k \in \mathbb{L}. \quad (5)$$

Proof: Since the first layer is not affected by other layers, system (1) restricted on R_{1i} is as follows:

$$\begin{aligned} dx_k(t) = & f_k(x_k(t), x_k(t - \tau), t)dt \\ & + \sum_{h \in V(R_{1i})} b_{kh}(t)H_{kh}(x_k(t), x_h(t))dt \\ & + g_k(x_k(t), x_k(t - \tau), t)dB(t), k \in V(R_{1i}). \end{aligned} \quad (6)$$

In [36] and [37], we know that some asymptotical properties of system (6) can be characterized by its limit system by the theory of asymptotically autonomous systems. Therefore, in order to study the stability of system (6), we only need to investigate the stability of its limit system. By Assumption 3, we know that the limit of $b_{kh}(t)$ exists, and then we give its limit system as follows:

$$\begin{aligned} dx_k(t) = & \tilde{f}_k(x_k(t), x_k(t - \tau))dt \\ & + \sum_{h \in V(R_{1i})} b_{kh}H_{kh}(x_k(t), x_h(t))dt \\ & + \tilde{g}_k(x_k(t), x_k(t - \tau))dB(t), \quad k \in V(R_{1i}) \end{aligned} \quad (7)$$

where $b_{kh}(t) \rightarrow b_{kh}$, as $t \rightarrow \infty$. Let

$$V_{1i}(x^{1i}, t) = \sum_{k \in V(R_{1i})} c_k^{1i} V_k(x_k, t)$$

where x^{1i} represents the state of the i th strongly connected component of the first layer, c_k^{1i} is the cofactor of the k th diagonal element of the Laplace matrix of (R_{1i}, B_{1i}) , $B_{1i} = (b_{kh})_{k,h \in V(R_{1i})}$. On account of the fact that R_{1i} is a strongly connected component, we have $c_k^{1i} > 0$ for any $k \in V(R_{1i})$. Denote $\beta^{1i} = \sum_{k \in V(R_{1i})} c_k^{1i} \beta_k$ and

$$\alpha^{1i} = \left(\sum_{k \in V(R_{1i})} c_k^{1i} \alpha_k \right)^{1-\frac{p}{2}} \left(\min_{k \in V(R_{1i})} \{c_k^{1i} \alpha_k\} \right)^{\frac{p}{2}} > 0$$

since $c_k^{1i} > 0$. In view of condition V1, we can get

$$V_{1i}(x^{1i}, t) \leq \sum_{k \in V(R_{1i})} c_k^{1i} \beta_k |x_k|^p \leq \beta^{1i} |x^{1i}|^p$$

and

$$\begin{aligned} V_{1i}(x^{1i}, t) & \geq \sum_{k \in V(R_{1i})} c_k^{1i} \alpha_k |x_k|^p \\ & = \sum_{m \in V(R_{1i})} c_m^{1i} \alpha_m \sum_{k \in V(R_{1i})} \left[\frac{c_k^{1i} \alpha_k}{\sum_{j \in V(R_{1i})} c_j^{1i} \alpha_j} (|x_k|^2)^{\frac{p}{2}} \right] \\ & \geq \sum_{m \in V(R_{1i})} c_m^{1i} \alpha_m \left[\sum_{k \in V(R_{1i})} \frac{c_k^{1i} \alpha_k}{\sum_{j \in V(R_{1i})} c_j^{1i} \alpha_j} |x_k|^2 \right]^{\frac{p}{2}} \\ & \geq \left(\sum_{k \in V(R_{1i})} c_k^{1i} \alpha_k \right)^{1-\frac{p}{2}} \left(\min_{k \in V(R_{1i})} \{c_k^{1i} \alpha_k\} \right)^{\frac{p}{2}} |x^{1i}|^p = \alpha^{1i} |x^{1i}|^p. \end{aligned}$$

Therefore, $\alpha^{1i} |x^{1i}|^p \leq V_{1i}(x^{1i}, t) \leq \beta^{1i} |x^{1i}|^p$. Fix any $\phi \in L^2_{\mathcal{F}_0}([-\tau, 0]; \mathbb{R}^m)$ and write $x(t, \phi) = x(t)$. By using (3),

we can derive that

$$\begin{aligned} \mathcal{L}V_{1i}(x^{1i}(t), t) & \leq \sum_{k \in V(R_{1i})} c_k^{1i} \left(\sum_{h \in V(R_{1i})} d_{kh} [P_h(x_h(t)) - P_k(x_k(t))] \right. \\ & \quad \left. - \sigma_k |x_k(t)|^p + \eta_k |x_k(t - \tau)|^p \right) \\ & = \sum_{k \in V(R_{1i})} \sum_{h \in V(R_{1i})} c_k^{1i} d_{kh} [P_h(x_h(t)) - P_k(x_k(t))] \\ & \quad - \sum_{k \in V(R_{1i})} c_k^{1i} \sigma_k |x_k(t)|^p + \sum_{k \in V(R_{1i})} c_k^{1i} \eta_k |x_k(t - \tau)|^p. \end{aligned}$$

We define $\mathbb{Q}_{1i}$ as the set of all spanning unicyclic graphs of (R_{1i}, B_{1i}) , and then a direct application of [19, Th. 2.2] yields

$$\begin{aligned} \sum_{k \in V(R_{1i})} \sum_{h \in V(R_{1i})} c_k^{1i} d_{kh} [P_h(x_h(t)) - P_k(x_k(t))] \\ = \sum_{\mathcal{Q} \in \mathbb{Q}_{1i}} W(\mathcal{Q}) \sum_{(s,r) \in E(\mathcal{Q})} [P_r(x_r(t)) - P_s(x_s(t))] = 0. \end{aligned}$$

Therefore, we have

$$\begin{aligned} \mathcal{L}V_{1i}(x^{1i}(t), t) \leq & - \sum_{k \in V(R_{1i})} c_k^{1i} \sigma_k |x_k(t)|^p \\ & + \sum_{k \in V(R_{1i})} c_k^{1i} \eta_k |x_k(t - \tau)|^p. \end{aligned} \quad (8)$$

By (5), there exists a sufficiently small $\gamma_{1i} > 0$, such that

$$(\gamma_{1i} \beta_k - \sigma_k) + e^{\gamma_{1i} \tau} \eta_k < 0. \quad (9)$$

Using Itô's formula and (8), one can get

$$\begin{aligned} \mathbb{E}[e^{\gamma_{1i} t} V_{1i}(x^{1i}(t), t)] & \leq \mathbb{E}V_{1i}(\phi(0), 0) + \mathbb{E} \int_0^t e^{\gamma_{1i} s} \\ & \quad \times \left[\gamma_{1i} \sum_{k \in V(R_{1i})} \beta_k c_k^{1i} |x_k(s)|^p \right. \\ & \quad \left. - \sum_{k \in V(R_{1i})} c_k^{1i} \sigma_k |x_k(s)|^p + \sum_{k \in V(R_{1i})} c_k^{1i} \eta_k |x_k(s - \tau)|^p \right] ds \\ & \leq \sum_{k \in V(R_{1i})} (\gamma_{1i} \beta_k - \sigma_k) \mathbb{E} \int_0^t e^{\gamma_{1i} s} c_k^{1i} |x_k(s)|^p ds \\ & \quad + e^{\gamma_{1i} \tau} \sum_{k \in V(R_{1i})} \eta_k \mathbb{E} \int_0^t e^{\gamma_{1i} s} c_k^{1i} |x_k(s)|^p ds \\ & \quad + e^{\gamma_{1i} \tau} \sum_{k \in V(R_{1i})} \eta_k \mathbb{E} \int_{-\tau}^0 e^{\gamma_{1i} s} c_k^{1i} |x_k(s)|^p ds + \mathbb{E}V_{1i}(\phi(0), 0). \end{aligned}$$

By (9), it follows that:

$$\begin{aligned} & \mathbb{E}[e^{\gamma_{1i}t} V_{1i}(x^{1i}(t), t)] \\ & \leq e^{\gamma_{1i}t} \sum_{k \in V(R_{1i})} \eta_k \mathbb{E} \int_{-\tau}^0 e^{\gamma_{1i}s} c_k^{1i} \max_{k \in V(R_{1i})} |x_k(s)|^p ds \\ & \quad + \beta^{1i} \mathbb{E} \|\phi\|^p \leq \beta^{1i} \mathbb{E} \|\phi\|^p \\ & \quad + \sum_{k \in V(R_{1i})} \eta_k e^{\gamma_{1i}t} \int_{-\tau}^0 e^{\gamma_{1i}s} c_k^{1i} ds \mathbb{E} \|\phi\|^p \triangleq C_{1i} \mathbb{E} \|\phi\|^p \end{aligned}$$

where $C_{1i} = \beta^{1i} + \sum_{k \in V(R_{1i})} \eta_k c_k^{1i} (e^{\gamma_{1i}\tau} - 1/\gamma_{1i})$. Hence, it yields that

$$\mathbb{E}[\alpha^{1i} e^{\gamma_{1i}t} |x^{1i}(t)|^p] \leq \mathbb{E}[e^{\gamma_{1i}t} V_{1i}(x^{1i}(t), t)] \leq C_{1i} \mathbb{E} \|\phi\|^p$$

and $\mathbb{E}|x^{1i}(t)|^p \leq (C_{1i}/\alpha^{1i}) e^{-\gamma_{1i}t} \mathbb{E} \|\phi\|^p$. Then, we can get

$$\begin{aligned} & \limsup_{t \rightarrow \infty} \frac{1}{t} \ln \mathbb{E}|x^{1i}(t)|^p \\ & \leq \limsup_{t \rightarrow \infty} \frac{1}{t} \left[-\gamma_{1i}t + \ln \left(\frac{C_{1i}}{\alpha^{1i}} \mathbb{E} \|\phi\|^p \right) \right] = -\gamma_{1i}. \end{aligned}$$

This implies that the p th moment Lyapunov exponent is not greater than $-\gamma_{1i}$. Similarly, we can verify that system (1) restricted on all strongly connected components in the first layer is ME-stable, so system (1) restricted on the first layer is ME-stable. Next, we investigate the stability of the trivial solution of system (1) restricted on R_{2i} . First, system (1) restricted on R_{2i} is as follows:

$$\begin{aligned} dx_k(t) = & \left[f_k(x_k(t), x_k(t-\tau), t) \right. \\ & + \sum_{h \in V(R_{2i})} b_{kh}(t) H_{kh}(x_k(t), x_h(t)) \\ & + \sum_{h \in \bigcup_j V(R_{1i})} c_{kh}(t) H_{kh}(x_k(t), x_h(t)) \left. \right] dt \\ & + g_k(x_k(t), x_k(t-\tau), t) dB(t), \quad k \in V(R_{2i}). \quad (10) \end{aligned}$$

Then, we substitute the trivial solution of system (6) to system (10) and in view of Assumption 2, the limit system of system (10) is as follows:

$$\begin{aligned} dx_k(t) = & \tilde{f}_k(x_k(t), x_k(t-\tau))dt \\ & + \sum_{h \in V(R_{2i})} b_{kh} H_{kh}(x_k(t), x_h(t))dt \\ & + \tilde{g}_k(x_k(t), x_k(t-\tau))dB(t), \quad k \in V(R_{2i}). \quad (11) \end{aligned}$$

Let $V_{2i}(x^{2i}, t) = \sum_{k \in V(R_{2i})} c_k^{2i} V_k(x_k, t)$, where x^{2i} represents the state of the i th strongly connected component in the second layer and c_k^{2i} is the cofactor of the k th diagonal element of the Laplace matrix of (R_{2i}, B_{2i}) , where $B_{2i} = (b_{kh})_{k,h \in V(R_{2i})}$. Denote $\beta^{2i} = \sum_{k \in V(R_{2i})} c_k^{2i} \beta_k$ and

$$\alpha^{2i} = \left(\sum_{k \in V(R_{2i})} c_k^{2i} \alpha_k \right)^{1-\frac{p}{2}} \left(\min_{k \in V(R_{2i})} \{c_k^{2i} \alpha_k\} \right)^{\frac{p}{2}}.$$

According to the proof of system (1) restricted on R_{1i} , it is easy to get $\alpha^{2i} |x^{2i}|^p \leq V_{2i}(x^{2i}, t) \leq \beta^{2i} |x^{2i}|^p$

$$\mathbb{E}[\alpha^{2i} e^{\gamma_{2i}t} |x^{2i}(t)|^p] \leq \mathbb{E}[e^{\gamma_{2i}t} V_{2i}(x^{2i}(t), t)] \leq C_{2i} \mathbb{E} \|\phi\|^p$$

and

$$\mathbb{E}|x^{2i}(t)|^p \leq \frac{C_{2i}}{\alpha^{2i}} e^{-\gamma_{2i}t} \mathbb{E} \|\phi\|^p.$$

Then, we have

$$\begin{aligned} & \limsup_{t \rightarrow \infty} \frac{1}{t} \ln \mathbb{E}|x^{2i}(t)|^p \\ & \leq \limsup_{t \rightarrow \infty} \frac{1}{t} \left[-\gamma_{2i}t + \ln \left(\frac{C_{2i}}{\alpha^{2i}} \mathbb{E} \|\phi\|^p \right) \right] = -\gamma_{2i}. \end{aligned}$$

Therefore, the trivial solution of system (1) restricted on R_{2i} is ME-stable. Repeating the above-mentioned process, we can verify that the trivial solution of system (1) restricted on every strongly connected component of each layer is ME-stable. Therefore, the trivial solution of system (1) is ME-stable. ■

Remark 3: Theorem 2 presents a sufficient criterion to guarantee the ME-stability of SCTGS. Currently, the study about coupled systems has received considerable interest, and several related results have been obtained [8], [16]–[19]. However, we note that the coupling structure is constant in these literatures. Coupled systems with a constant coupling structure may be not reasonable to explain some phenomena in practice. For example, the dispersal rate among species lived in different patches always changes over time in biomathematics. Therefore, they are special cases of coupled systems with the time-varying coupling structure, and the result we obtain here is the generalization of the results obtained in previous literatures.

Remark 4: Theorem 2 provides an effective method to investigate the stability of system (1) established on a DWSC. The method is combining the theory of asymptotically autonomous systems, graph theory, and Lyapunov method, which not only overcomes the difficulty of constructing an appropriate Lyapunov function directly but also facilitates the proof for the stability of system (1) established on a DWSC.

Since the previous results are based on vertex Lyapunov functions, vertex Lyapunov functions are not easy to construct sometimes. We shall establish a coefficients-type criterion for the ME-stability of system (1), and it can verify the stability of system (1) much more directly and be proved to be quite useful in practice.

B. Coefficients-Type Criterion

Theorem 3: The trivial solution of system (1) is ME-stable if Assumptions 1–3 are satisfied, and for any $k, h \in \mathbb{L}$, the following conditions hold.

B1: There are positive constants α_k , β_k , δ_k , and ϵ_k such that

$$x_k^T \tilde{f}_k(x_k, \tilde{x}_k) \leq -\alpha_k |x_k|^2 + \beta_k |\tilde{x}_k|^2$$

and

$$|\tilde{g}_k(x_k, \tilde{x}_k)|^2 \leq \delta_k |x_k|^2 + \epsilon_k |\tilde{x}_k|^2.$$

B2: There exists a constant A_{kh} such that

$$|H_{kh}(x_k, x_h)| \leq A_{kh} (|x_k| + |x_h|).$$

B3: Let $p \geq 2$. The following inequality holds:

$$-\sigma_k + \eta_k < 0$$

in which

$$\sigma_k = p\alpha_k - (p-2)\beta_k - p(p-1)\delta_k - (p-1)(p-2)\epsilon_k - p \sum_{h \in V(R_{ji})} b_{kh} A_{kh}, \quad \eta_k = 2(\beta_k + (p-1)\epsilon_k).$$

Proof: By the proof of Theorem 2, we know that the limit system of system (1) restricted on R_{ji} is as follows:

$$\begin{aligned} dx_k(t) &= \tilde{f}_k(x_k(t), x_k(t-\tau))dt \\ &+ \sum_{h \in V(R_{ji})} b_{kh} H_{kh}(x_k(t), x_h(t))dt \\ &+ \tilde{g}_k(x_k(t), x_k(t-\tau))dB(t), \quad k \in V(R_{ji}). \end{aligned}$$

Define $V_k(x_k, t) = |x_k|^p$, for any $k \in V(R_{ji})$. Obviously, condition V1 is satisfied. By virtue of conditions B1 and B2, the differential operator $\mathcal{L}V_k(x_k(t), t)$ can be computed as

$$\begin{aligned} \mathcal{L}V_k(x_k(t), t) &\leq p|x_k(t)|^{p-2}x_k^T(t)\tilde{f}_k(x_k(t), \tilde{x}_k(t)) \\ &+ p|x_k(t)|^{p-2}x_k^T(t) \sum_{h \in V(R_{ji})} b_{kh} H_{kh}(x_k(t), x_h(t)) \\ &+ p(p-1)|x_k(t)|^{p-2}|\tilde{g}_k(x_k(t), \tilde{x}_k(t))|^2 \\ &\leq p|x_k(t)|^{p-2}(-\alpha_k|x_k(t)|^2 + \beta_k|x_k(t-\tau)|^2) \\ &+ p(p-1)|x_k(t)|^{p-2}(\delta_k|x_k(t)|^2 + \epsilon_k|x_k(t-\tau)|^2) \\ &+ p|x_k(t)|^{p-2}x_k^T(t) \sum_{h \in V(R_{ji})} b_{kh} H_{kh}(x_k(t), x_h(t)) \\ &\leq 2(\beta_k + (p-1)\epsilon_k)|x_k(t-\tau)|^p \\ &+ (-p\alpha_k + (p-2)\beta_k \\ &+ p(p-1)\delta_k + (p-1)(p-2)\epsilon_k)|x_k(t)|^p \\ &+ p|x_k(t)|^{p-2}x_k^T(t) \sum_{h \in V(R_{ji})} b_{kh} H_{kh}(x_k(t), x_h(t)). \end{aligned}$$

By using Young's inequality

$$|a|^p|b|^q \leq \frac{p}{p+q}|a|^{p+q} + \frac{q}{p+q}|b|^{p+q}$$

it is easy to see that

$$\begin{aligned} p|x_k(t)|^{p-2}x_k^T(t) \sum_{h \in V(R_{ji})} b_{kh} H_{kh}(x_k(t), x_h(t)) \\ \leq p|x_k(t)|^{p-1} \sum_{h \in V(R_{ji})} b_{kh} A_{kh}(|x_k(t)| + |x_h(t)|) \\ \leq \sum_{h \in V(R_{ji})} b_{kh} A_{kh}[P_h(x_h(t)) - P_k(x_k(t))] \\ + p \sum_{h \in V(R_{ji})} b_{kh} A_{kh}|x_k(t)|^p \end{aligned}$$

where $P_h(x_h) = |x_h|^p$. Combining the computations above, we can get

$$\begin{aligned} \mathcal{L}V_k(x_k(t), t) &\leq \left(-p\alpha_k + (p-2)\beta_k + p(p-1)\delta_k \right. \\ &\quad \left. + (p-1)(p-2)\epsilon_k + p \sum_{h \in V(R_{ji})} b_{kh} A_{kh} \right) |x_k(t)|^p \end{aligned}$$

$$\begin{aligned} &+ 2(\beta_k + (p-1)\epsilon_k)|x_k(t-\tau)|^p \\ &+ \sum_{h \in V(R_{ji})} b_{kh} A_{kh}[P_h(x_h(t)) - P_k(x_k(t))] \\ &= -\sigma_k|x_k(t)|^p + \eta_k|x_k(t-\tau)|^p \\ &+ \sum_{h \in V(R_{ji})} d_{kh}[P_h(x_h(t)) - P_k(x_k(t))] \end{aligned}$$

where $d_{kh} = b_{kh} A_{kh}$. Therefore, condition V2 holds. We verify that the trivial solution of system (1) restricted on R_{ji} is ME-stable, which means that the trivial solution of system (1) is ME-stable. ■

Remark 5: If the digraph is strongly connected, then condition B3 can be reduced to

$$-p\alpha_k + (p-2)\beta_k + p(p-1)\delta_k + (p-1)(p-2)\epsilon_k + 2p \sum_{h=1}^l a_{kh}(t)A_{kh} \leq 0.$$

Remark 6: Coupled systems are applied to many fields extensively, such as neural networks. Actually, some nonautonomous networks can be characterized by coupled systems with the time-varying coupling structure. Therefore, Theorem 3 gives a new viewpoint to study the stability of nonautonomous neural networks.

Remark 7: The coefficients-type criterion means that the conditions are given in the form of coefficients of the system, which makes conditions in the criterion easy to be verified. Therefore, the sufficient criterion facilitates the stability analysis of system (1). In order to end this section, an example is given as an application in Section IV.

IV. APPLICATION TO SCTCS

On the DWSC $(\mathcal{G}, A(t))$, where $A(t) = (a_{kh}(t))_{l \times l}$ is the time-varying weighted matrix, we consider SCTCS as follows:

$$\begin{aligned} \ddot{x}_k(t) &+ \varphi_k(x_k(t))\dot{x}_k(t) + x_k(t) + \varepsilon_k x_k(t-\tau) \\ &+ \sum_{h=1}^l a_{kh}(t)H_{kh}(x_k(t), x_h(t)) = h_k(x_k(t))\dot{B}(t), \quad k \in \mathbb{L} \end{aligned} \quad (12)$$

where $\varepsilon_k > 0$, $H_{kh} : \mathbb{R}^1 \times \mathbb{R}^1 \rightarrow \mathbb{R}^1$ is a normalized interference function from vertex h to vertex k , and $h_k : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ stands for the random perturbation intense of the k th vertex.

Next, in order to investigate the stability of system (12), Theorem 4 is given.

Theorem 4: Assume that Assumptions 1–3 are satisfied. For any $k, h \in \mathbb{L}$, suppose that the following conditions hold.

C1: There are positive constants m_k, M_k, γ_k , and η_k , such that $m_k < |\varphi_k(x_k)| < M_k$ and $|h_k(x_k)| \leq \gamma_k|x_k|$.

C2: There is a constant A_{kh} such that

$$|H_{kh}(x_k, x_h)| \leq A_{kh}(|x_k| + |x_h|).$$

C3: There exists a constant μ_k such that $2\varepsilon_k < \mu_k \leq m_k$, $(2M_k - \mu_k)\mu_k \leq 4$ and

$$-\frac{\mu_k}{4} + \varepsilon_k + (p-1)\gamma_k^2 + \sum_{h \in V(R_{ji})} b_{kh} A_{kh} < 0.$$

Then the trivial solution of system (12) is ME-stable.

Proof: In view of conditions C1 and C2, we can obtain that system (12) exists a trivial solution. For any i, j , we consider the stability of the trivial solution of system (12) restricted on R_{ji} . Let $y_k(t) = \dot{x}_k(t) + \eta_k x_k(t)$, then system (12) restricted on R_{ji} can be rewritten as

$$\begin{cases} dx_k(t) = [y_k(t) - \eta_k x_k(t)]dt \\ dy_k(t) = \left[(\eta_k - \varphi_k(x_k(t)))y_k(t) - \varepsilon_k x_k(t - \tau) \right. \\ \quad \left. + (\eta_k \varphi_k(x_k(t)) - \eta_k^2 - 1)x_k(t) \right]dt \\ \quad - \left[\sum_{h \in V(R_{ji})} b_{kh}(t)H_{kh}(x_k(t), x_h(t)) \right. \\ \quad \left. + \sum_{h \in \bigcup_{j_1 < j} \bigcup_{i_1} V(R_{j_1 i_1})} c_{kh}(t)H_{kh}(x_k(t), x_h(t)) \right]dt \\ \quad \left. + h_k(x_k(t))dB(t), \quad k \in V(R_{ji}). \right] \end{cases} \quad (13)$$

By the proof of Theorem 2, we can find that the limit system of system (13) is given as follows:

$$\begin{cases} dx_k(t) = [y_k(t) - \eta_k x_k(t)]dt \\ dy_k(t) = \left[(\eta_k - \varphi_k(x_k(t)))y_k(t) - \varepsilon_k x_k(t - \tau) \right. \\ \quad \left. + (\eta_k \varphi_k(x_k(t)) - \eta_k^2 - 1)x_k(t) \right]dt \\ \quad - \sum_{h \in V(R_{ji})} b_{kh} H_{kh}(x_k(t), x_h(t))dt \\ \quad \left. + h_k(x_k(t))dB(t), \quad k \in V(R_{ji}). \right] \end{cases}$$

We define vector-value functions $X_k = (x_k, y_k)^T$, $\tilde{X}_k = (\tilde{x}_k, \tilde{y}_k)^T$ and

$$F_k(X_k, \tilde{X}_k, t) = \begin{pmatrix} y_k - \eta_k x_k \\ (\eta_k - \varphi_k(x_k))y_k + (\eta_k \varphi_k(x_k) - \eta_k^2 - 1)x_k - \varepsilon_k \tilde{x}_k \end{pmatrix}$$

thus

$$\tilde{F}_k(X_k, \tilde{X}_k) = \begin{pmatrix} y_k - \eta_k x_k \\ (\eta_k - \varphi_k(x_k))y_k + (\eta_k \varphi_k(x_k) - \eta_k^2 - 1)x_k - \varepsilon_k \tilde{x}_k \end{pmatrix}.$$

Then, according to condition C1, we can derive that

$$\begin{aligned} X_k^T \tilde{F}_k(X_k, \tilde{X}_k) &\leq (\eta_k \varphi_k(x_k) - \eta_k^2) |x_k y_k| - \eta_k x_k^2 \\ &\quad + (\eta_k - \varphi_k(x_k)) y_k^2 - \varepsilon_k \tilde{x}_k y_k \\ &\leq - \left(\varphi_k(x_k) - \eta_k - \frac{1}{2\lambda_k^2} (\eta_k \varphi_k(x_k) - \eta_k^2) - \frac{\varepsilon_k}{2} \right) y_k^2 \\ &\quad - \left(\eta_k - \frac{1}{2} (\eta_k \varphi_k(x_k) - \eta_k^2) \lambda_k^2 \right) x_k^2 + \frac{\varepsilon_k}{2} \tilde{x}_k^2 \\ &= - \left(\varphi_k(x_k) - \eta_k - \frac{1}{2\lambda_k^2} \eta_k (\varphi_k(x_k) - \eta_k) - \frac{\varepsilon_k}{2} \right) y_k^2 \\ &\quad - \eta_k \left(1 - \frac{1}{2} (\varphi_k(x_k) - \eta_k) \lambda_k^2 \right) x_k^2 + \frac{\varepsilon_k}{2} \tilde{x}_k^2. \end{aligned}$$

Let $\lambda_k^2 = \eta_k = (\mu_k/2)$, then we have $|h_k|^2 \leq \gamma_k^2 |x_k|^2$

$$X_k^T \tilde{F}_k(X_k, \tilde{X}_k) \leq - \left(\frac{\mu_k}{4} - \frac{\varepsilon_k}{2} \right) |X_k|^2 + \frac{\varepsilon_k}{2} |\tilde{X}_k|^2.$$

Therefore, condition B1 is satisfied. By condition C2, it is easy to see that condition B2 is satisfied. According to condition C3, we can get

$$-\sigma_k + \eta_k = - \left(\frac{\mu_k}{4} - \frac{\varepsilon_k}{2} \right) + (p-1) \gamma_k^2 + \frac{\varepsilon_k}{2} + \sum_{h \in V(R_{ji})} b_{kh} A_{kh} < 0$$

which implies that condition B3 is satisfied. We show that all conditions in Theorem 3 are satisfied. Therefore, by Theorem 3, we can get that the trivial solution of system (12) is ME-stable. Thus, the proof is complete. ■

Remark 8: Recently, the stability of coupled oscillators has been investigated by some researchers in [17], [19], and [20], and they have established coupled oscillators on a strongly connected digraph. In this paper, we investigate the stability of coupled oscillators on DWSCs. Therefore, the results we obtained are more general, which means that we generalize the results of our previous works.

V. NUMERICAL TEST

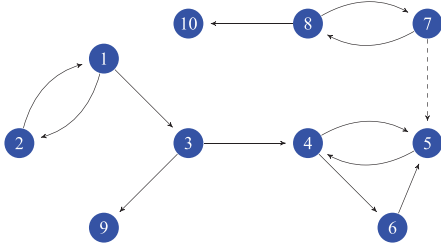
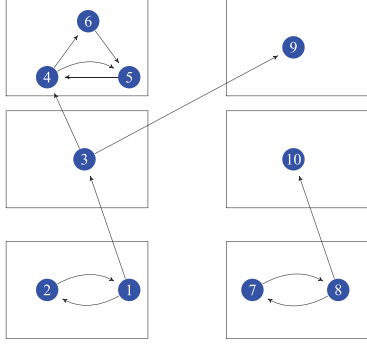
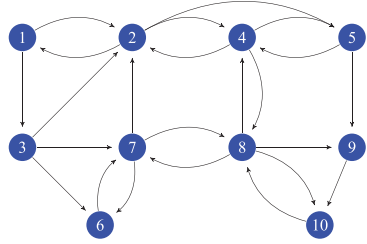
In this section, we present two numerical examples to illustrate the effectiveness of our results in this paper.

Example 2: Let $l = 10$, and consider SCTCS in the form of system (12). We choose functions $\varphi_k(x_k) = 0.5 + 0.2 \cos x_k$, $H_{kh}(x_k, x_h) = 0.01 x_h$ and $h_k(x_k) = 0.03 \sin x_k$. Next, we give a DWSC $(\mathcal{G}, A(t))$. We remove the arc from vertex 7 to vertex 5 when $\lim_{t \rightarrow \infty} a_{57}(t) = 0$, then we obtain a digraph $(\mathcal{G}, \tilde{A}(t))$, which is the spanning subgraph of $(\mathcal{G}, A(t))$, $\tilde{A}(t) = (\tilde{a}_{kh}(t))_{10 \times 10}$, where $\tilde{a}_{12}(t) = e^{-t^2} + 0.1$, $\tilde{a}_{21}(t) = e^{-t} + 0.2$, $\tilde{a}_{31}(t) = \sin t + 1.1$, $\tilde{a}_{43}(t) = \cos t + 1.2$, $\tilde{a}_{45}(t) = e^{-t} + 0.2$, $\tilde{a}_{54}(t) = e^{-t^2} + 0.3$, $\tilde{a}_{56}(t) = 2e^{-t} + 0.1$, $\tilde{a}_{64}(t) = e^{-t^2} + 0.2$, $\tilde{a}_{7,8}(t) = e^{-2t} + 0.2$, $\tilde{a}_{87}(t) = e^{-t} + 0.1$, $\tilde{a}_{93}(t) = \sin t + 1.3$, $\tilde{a}_{10,8}(t) = 0.1 \sin t + 0.3$. In the sequel, the digraph $(\mathcal{G}, \tilde{A}(t))$ can be condensed to a condensed digraph $\mathcal{R}$ (see Figs. 3 and 4). Obviously, Assumptions 1–3 are satisfied. Furthermore, we choose parameters as $\tau = 1$, $\eta_1 = 0.25$, $\eta_2 = 0.08$, $\eta_3 = 0.07$, $\eta_4 = 0.1$, $\eta_5 = 0.15$, $\eta_6 = 0.04$, $\eta_7 = 0.2$, $\eta_8 = 0.05$, $\eta_9 = 0.07$, $\eta_{10} = 0.25$, $\gamma_k = 0.03$, $M_k = 0.8$, $m_k = 0.3$, $\mu_k = 0.1$, $p = 2$, and $\varepsilon_k = 0.02$. We have

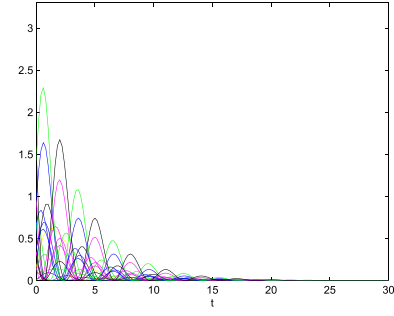
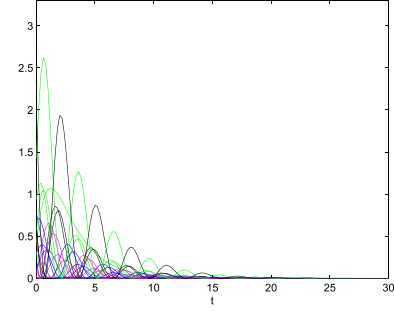
$$\max_{k \in \{1, \dots, 10\}} \left\{ -\frac{\mu_k}{4} + \varepsilon_k + (p-1) \gamma_k^2 + \sum_{h \in V(R_{ji})} b_{kh} A_{kh} \right\} < 0$$

where $A_{kh} = 0.01$, $2\varepsilon_k < \mu_k \leq m_k$, and $(2M_k - \mu_k)\mu_k \leq 4$. Therefore, all conditions in Theorem 4 are checked. Moreover, the initial conditions are selected as $x_1(t) = 0.4 \sin t - 0.7$, $y_1(t) = 1.2 \sin t \cos t - 0.3$, $x_2(t) = -0.3 \sin t \cos t + 0.5$, $y_2(t) = 0.4 \sin t \cos t + 0.7$, $x_3(t) = 1.2 \sin t - 0.5$, $y_3(t) = \cos t - 0.3 \sin t + 0.5$, $x_4(t) = 0.3 \sin t + 0.8 \cos t + 0.4$, $y_4(t) = -0.6 \sin t + 0.1$, $x_5(t) = 0.3 \sin t + 0.7$, $y_5(t) = \sin t + 0.4 \cos t - 0.1$, $x_6(t) = -0.7 \sin t + 0.7$, $y_6(t) = 0.4 \sin t - 0.2 \cos t + 0.35$, $x_7(t) = 0.8 \sin t + 0.5 \cos t - 0.3$, $y_7(t) = \cos t - 0.3 \sin t + 0.5$, $x_8(t) = 0.3 \cos t - 0.6 \sin t + 0.8$, $y_8(t) = -0.6 \sin t + 1.1$, $x_9(t) = -0.4 \sin t + 1.2$, $y_9(t) = \sin t + 0.2 \cos t - 0.6$, $x_{10}(t) = -0.2 \sin t + 0.5 \cos t + 1.2$, and $y_{10}(t) = 0.5 \sin t + 0.2 \cos t + 0.7$, where $t \in [-1, 0]$.

In addition, one considers system (12) on a strongly connected digraph $(\mathcal{G}, \tilde{B}(t))$, where $\tilde{B}(t) = (\tilde{b}_{kh}(t))_{10 \times 10}$,

Fig. 3. Digraph $(\mathcal{G}, A(t))$ with ten vertices, which is not strongly connected.Fig. 4. Condensed digraph $\mathcal{R}$ with three layers; each group is a strongly connected component.Fig. 5. Strongly connected digraph $(\mathcal{G}, \tilde{B}(t))$ with ten vertices.

$\tilde{b}_{12}(t) = \tilde{b}_{36}(t) = \tilde{b}_{54}(t) = \tilde{b}_{59}(t) = \tilde{b}_{76}(t) = \tilde{b}_{89}(t) = e^{-t^2}$, $\tilde{b}_{13}(t) = \tilde{b}_{21}(t) = \tilde{b}_{45}(t) = \tilde{b}_{52}(t) = \tilde{b}_{72}(t) = \tilde{b}_{87}(t) = \tilde{b}_{9,10}(t) = \tilde{b}_{10,8}(t) = e^{-t}$, $\tilde{b}_{24}(t) = \tilde{b}_{42}(t) = e^{-\cos t - t}$, $\tilde{b}_{32}(t) = \tilde{b}_{37}(t) = \tilde{b}_{48}(t) = \tilde{b}_{78}(t) = e^{-2t}$, $\tilde{b}_{67}(t) = e^{\sin t - t}$, $\tilde{b}_{84}(t) = e^{-2t^2}$, and $\tilde{b}_{8,10}(t) = e^{-t^2+1}$. We give a digraph $(\mathcal{G}, \tilde{B}(t))$ in Fig. 5. Moreover, we choose the following initial conditions: $x_1(t) = 0.7 \cos t - 0.9$, $y_1(t) = 1.2 \cos t - 0.1$, $x_2(t) = -0.6 \sin t + 0.4$, $y_2(t) = 0.2 \sin t \cos t + 0.5$, $x_3(t) = 0.7 \cos t - 0.8$, $y_3(t) = 0.3 \cos t - \sin t + 0.5$, $x_4(t) = 0.3 \sin t + 0.8 \cos t + 0.4$, $y_4(t) = -0.6 \cos t + 0.7$, $x_5(t) = 0.5 \cos t + 0.7$, $y_5(t) = 0.1 \sin t + 0.2 \cos t - 0.4$, $x_6(t) = -0.7 \cos t + 0.9$, $y_6(t) = 0.4 \sin t - 0.2 \cos t + 0.5$, $x_7(t) = 0.5 \sin t + 0.7 \cos t - 0.6$, $y_7(t) = 0.2 \cos t - 0.3 \sin t + 0.2$, $x_8(t) = 0.3 \cos t - 0.6 \sin t + 1.2$, $y_8(t) = -0.8 \cos t + 1.1$, $x_9(t) = -0.2 \cos t + 0.9$, $y_9(t) = 0.2 \sin t + \cos t - 0.4$, $x_{10}(t) = -0.2 \sin t + 0.5 \cos t + 1.2$, and $y_{10}(t) = 0.5 \sin t + 0.2 \cos t + 0.9$, where $t \in [-1, 0]$. Furthermore, the mean square figures of system (12) are presented in Figs. 6 and 7. Actually, we can see obviously from Figs. 6 and 7 that the trivial solution of system (12) is ME-stable, which implies that our results for strongly connected digraphs are also effective. Thus the numerical result shows the effectiveness and feasibility of the developed results.

Fig. 6. Mean square figure of system (12) restricted on DWSC $(\mathcal{G}, \tilde{A}(t))$.Fig. 7. Mean square figure of system (12) restricted on digraph $(\mathcal{G}, \tilde{B}(t))$ with strong connectivity.

Remark 9: In Section V, we establish a system on digraphs with strong connectedness and without strong connectedness to exhibit the result of Theorem 4 by a simulation, respectively, and then, two figures are obtained, which show that the trivial solution of system (12) is stable. It is easy to see that digraphs with strong connectedness also satisfy Theorem 4, which implies that digraphs without strong connectedness are more general. In addition, compared with the results in [8], [16], [20], and [39] we remove the assumption that digraphs should be strongly connected and simplify conditions in theorems; thus, our results generalize previous results.

Example 3: Stochastic neural networks are extensively investigated by many researchers, since their wide applications in practice. Huang *et al.* [38] considered a stochastic recurrent neural network as follows:

$$dx(t) = [-Cx(t) + AF(x(t)) + BG(x(t - \tau(t))) + J]dt + \sigma(x(t), x(t - \tau(t)), t)dB(t). \quad (14)$$

They investigated the p th moment stability of system (14). As a numerical example in this paper, we consider a special case of system (14) as

$$dx(t) = [-Cx(t) + A(t)F(x(t))]dt + \sigma(x(t), x(t - \tau), t)dB(t) \quad (15)$$

where

$$\sigma(x(t), x(t - \tau), t) = \text{diag}(\varrho_1, \varrho_2, \dots, \varrho_l)(g_1, g_2, \dots, g_l)^T$$

in which $g_i = g_i(x(t), x(t - \tau), t) \in \mathbb{R}^1$. Take $l = 6$, $C = \text{diag}(c_1, \dots, c_6) = \text{diag}(0.8, 1, 2, 2.4, 1.4, 1.8)$, $\tau = 1$, $A(t) = (a_{kh}(t))_{6 \times 6}$, where $a_{12}(t) = e^{-t} + 0.1$, $a_{21}(t) = e^{-t} + 0.3$, $a_{31}(t) = 0.01 \sin t + 1$, $a_{43}(t) = 0.02 \sin t + 1.2$, $a_{45}(t) = e^{-2t} + 0.2$, $a_{54}(t) = e^{-t} + 0.1$, $a_{53}(t) = e^{-t}$, $a_{56}(t) = e^{-t} + 0.5$, $a_{64}(t) = 2e^{-t} + 0.4$, then a DWSC is given

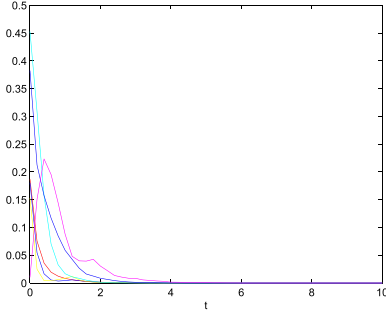
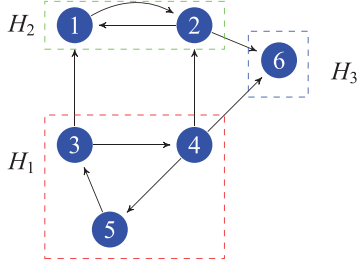


Fig. 8. Mean square figure of the trivial solution of system (15).

Fig. 9. Digraph $(\mathcal{G}, \tilde{D}(t))$ with six vertices, which is not strongly connected.

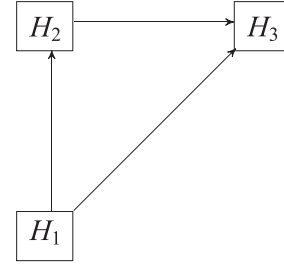
as $(\mathcal{G}, A(t))$. Removing the arc from vertex 3 to vertex 5 since $\lim_{t \rightarrow \infty} a_{53}(t) = 0$, one can get the spanning subgraph of $(\mathcal{G}, A(t))$ as $(\mathcal{G}, \tilde{A}(t))$. Next, we choose functions $F(x(t)) = (\tanh x_1(t), \dots, \tanh x_6(t))^T$ and $\sigma(x(t), x(t - \tau), t) = \text{diag}(0.1, 0.3, 0.5, 0.2, 0.4, 0.7)(x_1(t - \tau), \dots, x_6(t - \tau))^T$. Obviously, Assumptions 1–3 are satisfied. According to parameters and functions selected above, we can derive that $x_k^T \tilde{f}_k(x_k, \tilde{x}_k) = -c_k |x_k|^2$ and $|\tilde{g}_k(x_k, \tilde{x}_k)|^2 = \varrho_k^2 |\tilde{x}_k|^2$. Therefore, condition B1 holds. Moreover, $|H_{kh}(x_k, x_h)| = |\tanh x_h| \leq |x_h|$, so condition B2 holds. Let $p = 2$, then one can get $-\sigma_k + \eta_k = -c_k + 2 \sum_{h=1}^6 b_{kh} + 2\varrho_k < 0$, $k = 1, 2, \dots, 6$, where b_{kh} is the limit of $a_{kh}(t)$, and thus, condition B3 holds. Therefore, all conditions in Theorem 3 are verified. Furthermore, select initial conditions as $x_1(t) = 0.2 \cos t \sin t - 0.5$, $x_2(t) = 0.7 \sin t + 0.2 \cos t - 0.3$, $x_3(t) = -0.3 \cos t + 0.5$, $x_4(t) = 0.4 \sin t + 0.3$, $x_5(t) = 1.2 \sin t - 0.5$, and $x_6(t) = 0.5 \cos t - 0.3 \sin t + 0.2$, where $t \in [-1, 0]$. Then the mean square figure of system (15) is shown in Fig. 8. It is obvious that the trivial solution of system (15) is ME-stable, which means that our theoretical results are effective and feasible.

APPENDIX

In this section, in order to visually display the hierarchical method for readers, a digraph $(\mathcal{G}, \tilde{D}(t))$ with six vertices in Fig. 9 is taken for example.

Step 1: According to the hierarchical approach presented in Section II-A, the adjacency matrix of $(\mathcal{G}, \tilde{D}(t))$ is given first and defined as J as follows:

$$J = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

Fig. 10. Corresponding condensed digraph $\mathcal{R}$ of digraph $(\mathcal{G}, \tilde{D}(t))$.

Step 2: All strongly connected components are obtained by using Tarjan's strongly connected component algorithm, and Array1 and Array2 are acquired as $\text{Array1} = [5 \ 3 \ 4 \ 2 \ 1 \ 6]$, $\text{Array2} = [3 \ 2 \ 1]$.

Step 3: First, we have to alter the matrix J according to Array1, and then change the position of each row such that the order of each row is in accordance with that of each vertex in Array1 and it is the same with each line according to Array1. Finally, we acquire a new matrix defined as J' presented in the following:

$$J' = \left(\begin{array}{ccc|ccc} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{array} \right).$$

Step 4: We divide the matrix J' according to Array2. An example is given to illustrate how we can employ this by MATLAB. Divide J' using the function `mat2cell()` and corresponding parameters are $(J', \text{Array2}[], \text{Array2}[])$. Then, nine matrix blocks are obtained as follows:

$$\begin{aligned} J'_{11} &= \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad J'_{12} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad J'_{13} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ J'_{21} &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J'_{22} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J'_{23} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ J'_{31} &= (0 \ 0 \ 1), \quad J'_{32} = (1 \ 0), \quad J'_{33} = (0). \end{aligned}$$

Step 5: Each matrix block is dealt with the rule described in (5). Then a new matrix is acquired and defined as J'' , which is exactly the adjacency matrix of the condensed digraph according to Theorem 1 in the following:

$$J'' = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

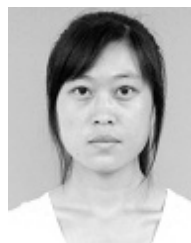
Step 6: First, seek for the rows whose elements are all zero and corresponding vertices belong to the first layer. We note that the first row satisfies the condition merely; therefore, the first layer contains vertex H_1 . Then, remove the first row and first line from matrix J'' and we see that the first row satisfies the condition, which implies that vertex H_2 belongs to the second layer. Then, it is the same with the rest of the vertices and we can obtain that the third layer contains the vertex H_3 obviously. The detail is displayed in Fig. 10.

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